

# Project Proposal

Course No: CSE 4122

## Interpretable Sentiment Classification in Software Engineering Communication

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**Problem.** Sentiment models trained on general text misclassify software engineering (SE) communication, where technical vocabulary (“fatal error,” “kill the process,” “crashed”) sounds negative but is functionally neutral. This domain shift reduces the reliability of sentiment tools applied to developer platforms (StackOverflow, GitHub, code review).

**Objective.** Build a sentiment classifier for SE text and use explainability methods to audit *why* it succeeds or fails on domain-specific vocabulary — not just measure accuracy.

**Dataset.** Senti4SD — ~4,000 StackOverflow posts labeled positive/negative/neutral (publicly available, `collab-uniba/Senti4SD`).

### Methodology.

- **Features:** TF-IDF with N-grams (1,2)/(1,3) to capture compound technical phrases; pretrained (GloVe) vs. self-trained (Word2Vec) embeddings for comparison.
- **Models:** Naive Bayes (generative) vs. Logistic Regression/SVM (discriminative), evaluated across all feature sets.
- **Explainability:** LIME/SHAP applied per-instance and aggregated across the test set to detect lexical bias; a hand-built stress-test set (technically harsh-but-neutral vs. genuinely negative sentences) evaluates whether models confuse jargon with hostility.
- **Evaluation:** Precision/Recall/F1 per class, confusion matrix (focus on negative-vs-neutral confusion).
- **Demo:** Notebook/CLI — input text, output prediction + highlighted contributing words.

### Expected Outcomes.

1. Comparative results on which representation (TF-IDF, pretrained, or domain-trained embeddings) best handles SE-specific vocabulary.
2. Generative vs. discriminative performance comparison.
3. Empirical evidence on whether models rely on domain-appropriate cues or general-language bias.
4. A working, interpretable demonstration pipeline.

**Limitations.** Small corpus size may limit self-trained embedding quality; explanation methods (LIME/SHAP) are approximations with known stability limitations, acknowledged in analysis.